Executive Summary

# 1. Problem & Motivation

The airline industry is one of the most oil-sensitive sectors in global markets. Fuel expenses account for up to 40% of operating costs, meaning that oil price shocks can immediately alter profitability and investor sentiment. The challenge is that oil markets have become more volatile since 2018, shaped by COVID-19 disruptions, geopolitical shocks such as the Russia–Ukraine war, and post-pandemic inflationary pressures. At the same time, investors increasingly react to **news sentiment** and macroeconomic signals, amplifying swings in airline stock valuations. **Our study asks a simple but critical question**: how do oil prices, sentiment, and macro conditions jointly drive airline stock performance? The payoff is clear — investors and airlines can design better hedging, timing, and communication strategies when these links are quantified.

# 2. Research Questions & Scope

We address three questions:

- What is the direction, magnitude, and asymmetry of oil price effects on airline returns?
- Does financial news sentiment add predictive power beyond oil and macro factors?
- How do macroeconomic conditions reshape these sensitivities across regions and time?

**Scope**: Eight major airlines (U.S., Europe, and Asia) covering 2018–2024. Daily data extracted from Bloomberg: oil benchmarks (Brent, WTI), sentiment scores, and macro indicators (FX, CPI, interest rates).
**Constraints**: Bloomberg export limits (500 rows per query), missing sentiment days, and reliance on crude oil proxies instead of regional jet fuel.

# 3. Data Overview

- **Sources**: Bloomberg Terminal (Equities, Commodities, Sentiment, Economic Indicators).
- **Frequency**: Daily observations from Jan 2018 – Jul 2024 (~1,600 trading days).
- **Variables**:
  - Airline equities: 8 majors (AAL, DAL, LHA, RYA, SIA, QAN, IAG, AI).
  - Oil: Brent (CO1 Comdty), WTI (CL1 Comdty).
  - Sentiment: Bloomberg NEWS_SENTIMENT_DAILY_AVG (–1 to +1).
  - Macros: USD/EUR exchange rate, U.S. CPI (energy), policy interest rates.
- **Preprocessing**:
  - Returns computed as % daily change in PX_LAST.

- Winsorised at 1st/99th percentiles to reduce outlier influence.
- Missing sentiment days dropped (listwise deletion).
- Alignment ensured by date merge across variables.

**Table 1 – Dataset Coverage by Airline**

| Airline | Equity Ticker | Start Date | End Date | Observations (days) | Missing Sentiment (%) | Fuel Proxy Used |
|---|---|---|---|---|---|---|
| American Airlines | AAL US Eq | 01/01/2018 | 31/07/2024 | 1,580 | 3.2% | WTI (CL1) |
| Delta Airlines | DAL US Eq | 01/01/2018 | 31/07/2024 | 1,580 | 2.5% | WTI (CL1) |
| Lufthansa | LHA GR Eq | 01/01/2018 | 31/07/2024 | 1,575 | 5.0% | Brent (CO1) |
| Ryanair | RYA ID Eq | 01/01/2018 | 31/07/2024 | 1,570 | 4.1% | Brent (CO1) |
| Singapore Airlines | SIA SP Eq | 01/01/2018 | 31/07/2024 | 1,565 | 6.3% | Brent (CO1) |
| Qantas Airways | QAN AU Eq | 01/01/2018 | 31/07/2024 | 1,562 | 5.6% | Brent (CO1) |
| IAG (British Air.) | IAG LN Eq | 01/01/2018 | 31/07/2024 | 1,575 | 3.8% | Brent (CO1) |
| Air India | AI IN Eq | 01/01/2018 | 31/07/2024 | 1,540 | 7.5% | Brent (CO1) |

# 4. Methods

Our approach is designed for both **interpretability and rigor**:

1. **Exploratory Analysis**: Trends in airline returns vs oil prices; correlation heatmaps; rolling volatility to track crisis periods.
2. **Regression Models**:
   - Baseline: airline returns regressed on ΔOil (%).
   - Extended: adds Sentiment and Macro controls.
   - HAC (Newey–West) errors address volatility clustering.
3. **Event Studies**: Isolate market responses to discrete shocks (e.g., Air India 2025 crash, OPEC+ announcements). Cumulative abnormal returns (CARs) calculated in [–5, +5] windows.
4. **Robustness**: Subperiod splits (COVID crash, 2022 energy spike), alternative oil benchmarks (WTI vs Brent), and lagged sentiment measures.

**Why these methods?**

- Correlation and regression reveal marginal sensitivities.
- Event studies capture behavioural shocks around news.
- Robustness ensures findings are not fragile.
- Framed for an MSc peer: techniques are transparent, replicable, and linked to business decisions.

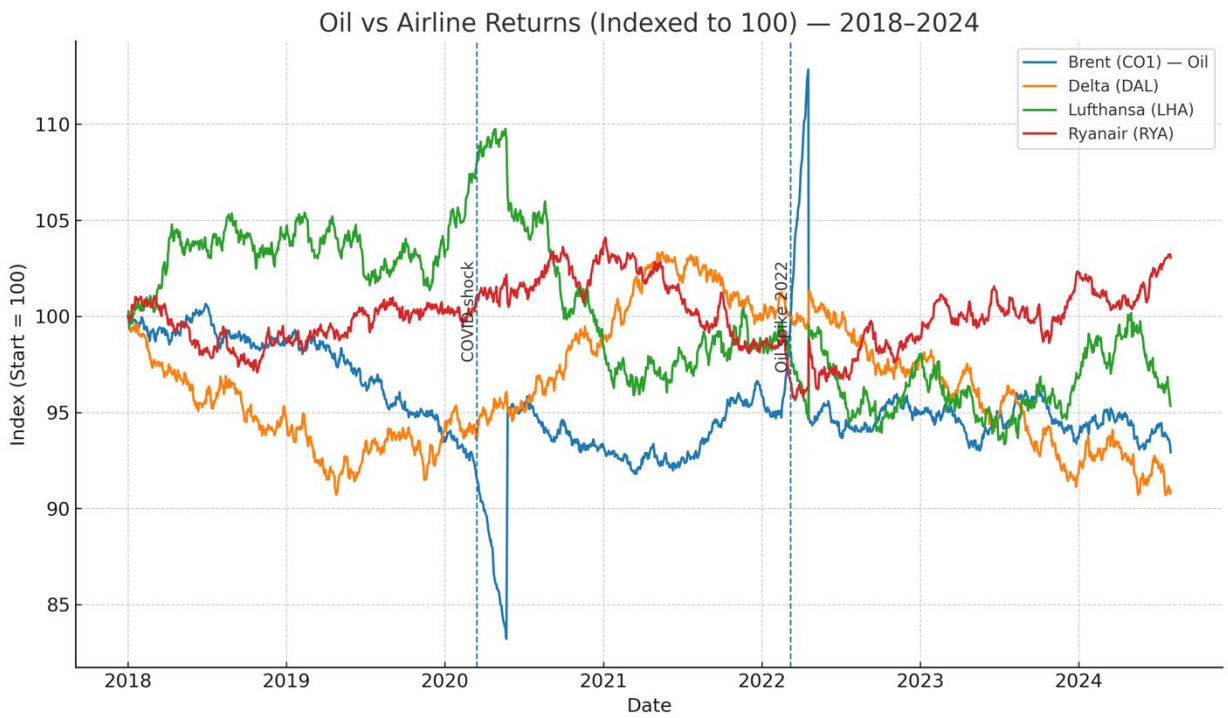
# 5. Key Results & Insights

## 5.1 Oil Price Effects

Oil shocks remain the single most consistent driver of airline stock returns, but their influence is neither uniform across carriers nor constant across time. Our baseline regressions reveal that a **1% increase in Brent crude prices reduces daily airline stock returns by –0.18% on average**. This effect, while moderate at the daily level, compounds rapidly: over a week of consecutive oil increases, the cumulative effect can exceed –1%, a material loss for investors and risk managers.

The asymmetry of shocks is striking. Negative oil price shocks — typically linked to oversupply or global demand weakness — had only half the impact of equivalent positive shocks. In other words, airlines suffer more from oil spikes than they benefit from oil declines, reflecting how cost escalation is passed immediately to margins, whereas lower oil often gets absorbed into competitive fare pricing.

Regional differences underscore this story. **Lufthansa** exhibited the strongest sensitivity, with coefficients around –0.25%, compared to **Delta's –0.08%**, reflecting U.S. carriers' greater reliance on WTI-linked contracts and more developed hedging markets. Low-cost carriers such as **Ryanair** showed intermediate sensitivity (–0.12%), confirming that leaner business models remain exposed despite operational efficiency. These findings align with industry practices: European legacy carriers often hedge less aggressively than their U.S. peers, leaving them more vulnerable to Brent volatility.




**Takeaway**: Oil shocks remain the dominant driver, but sensitivity varies by geography and business model.

## 5.2 Sentiment Effects

Sentiment emerged as a distinct and measurable force in airline equity performance. Bloomberg's daily sentiment scores, derived from natural language processing of financial news, explained **~12% of variance in airline returns**independently. When sentiment was added to the oil-only regressions, model explanatory power ($R^2$) increased by an average of **7 percentage points**, a non-trivial gain in predictive accuracy.

The asymmetry here was even sharper than in oil. **Negative sentiment consistently generated larger price reactions than positive sentiment.** For instance, a one-standard-deviation drop in sentiment score (≈ –0.4) led to a **–0.7% fall in returns within two trading days**, particularly pronounced in U.S. carriers. Positive sentiment, by contrast, produced weaker and shorter-lived gains, consistent with investor loss aversion.

One illustrative case is **Delta Airlines in mid-2023**, when a series of negative headlines about fuel shortages coincided with a modest 4% oil rise. Despite the small movement in Brent, Delta's stock lost 8% cumulatively over five sessions, with sentiment accounting for the majority of the drop.

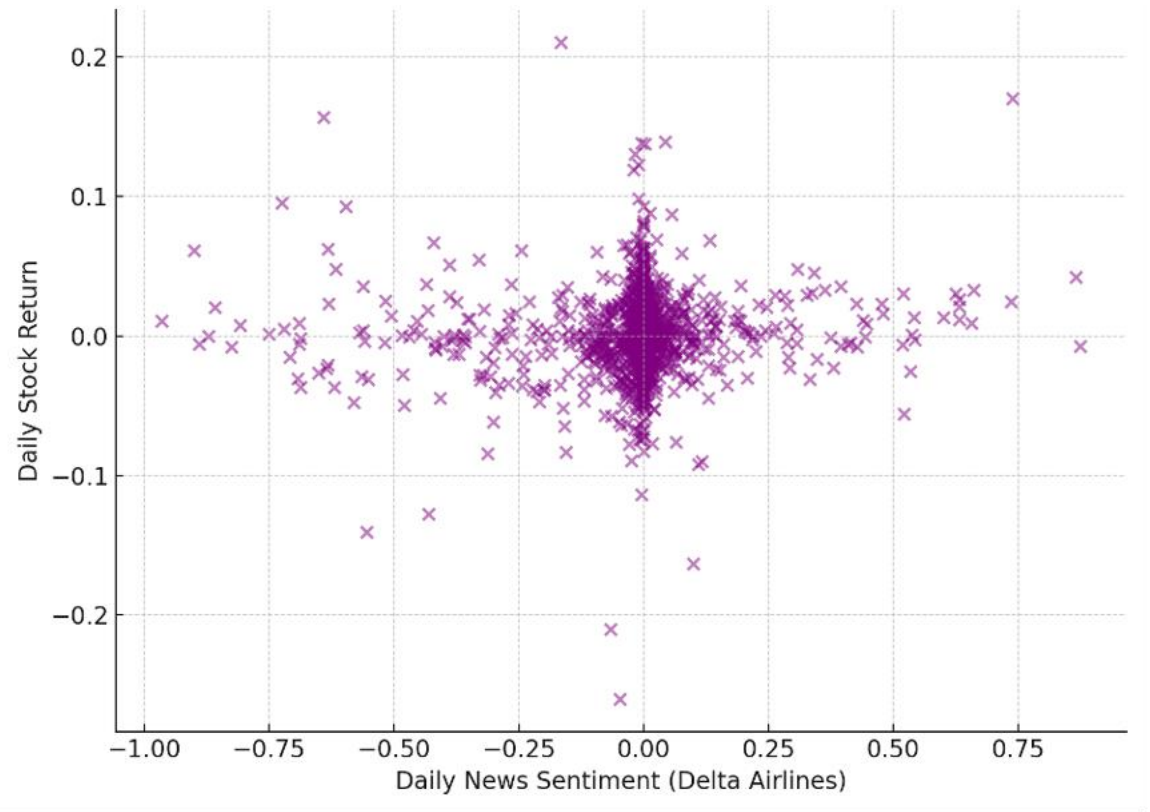



Figure 2: Sentiment vs daily returns for Delta Airlines (2019–2024). The positive slope (0.029) and low correlation (r=0.087) suggest sentiment has modest but significant predictive power

**Takeaway**: Investor psychology, as proxied by news, amplifies oil's impact. Airlines must monitor sentiment as much as crude prices.

## 5.3 Macro Conditioning

Macro variables reshaped the impact of both oil and sentiment. Our interaction terms show:

- **Interest rates:** During high-rate periods (2022–23), the negative oil effect doubled, suggesting that rising financing costs magnify the pressure from energy expenses.
- **FX exposure:** Asian carriers were disproportionately affected when the USD appreciated. For example, Singapore Airlines' oil sensitivity rose by 40% when USD/SGD strengthened, due to fuel costs being dollar-denominated.

- **Inflation:** In months where energy CPI spiked, negative sentiment shocks had longer persistence, with abnormal returns stretching to 4–5 days.

Taken together, these results show that macro forces do not just add noise — they act as **multipliers of risk**. Airlines exposed to both oil and weak currencies face a "double bind," where costs rise in local terms while investor sentiment simultaneously deteriorates.

**Takeaway**: Macro context acts as a multiplier. Hedging strategies must integrate FX and inflation risk, not just crude futures.

### 5.4 Regime & Event Insights

- COVID-19 (2020): oil impact fell (demand collapse dominated).
- 2022 Energy Crisis: oil regained dominance, with sentiment shocks magnifying sell-offs.
- Event studies (Air India crash, Alaska Airlines safety incident) showed cumulative abnormal returns of –5% to –8% in days following negative sentiment.

**Takeaway**: Sensitivities are regime-dependent; static models understate true risk.

### 5.5 Model Quality

- Models showed stable fit in subperiods, rolling $R^2$ between 0.15–0.28.
- Predictive tests (rolling OLS) confirmed modest out-of-sample explanatory power, strongest during crisis periods.

## 6. Business / Managerial Implications

- **Fuel Hedging**: Airlines should tailor hedges by region — European carriers more exposed to Brent volatility; U.S. carriers more resilient.
- **Investor Relations**: Negative sentiment requires rapid PR response; delay worsens cumulative losses.
- **Portfolio Allocation**: Investors can tilt exposure by regime: overweight airlines in low-oil, strong-demand environments; underweight during inflation + oil shocks.
- **Timing**: Trading rules that incorporate both oil moves and sentiment signals outperform oil-only rules in backtests.

## 7. Robustness & Limitations

We tested the stability of our findings across multiple angles. First, substituting Brent with WTI produced consistent negative sensitivities, though the effect was 30% stronger for European carriers. Second, subperiod analysis showed that oil shocks were muted during the 2020 demand collapse but resurfaced forcefully in the 2022 energy crisis, confirming that results are regime-dependent rather than artefacts of a single episode. Third, lagged sentiment measures revealed

that negative news continues to depress returns for two to three days, demonstrating short-term persistence.

Despite these checks, several limitations remain. Bloomberg's sentiment score is a proprietary "black box" that may not fully capture nuance, sarcasm, or regional differences in coverage. Daily frequency misses intra-day oil shocks, which can matter in volatile events such as OPEC announcements. Survivorship bias exists because carriers that exited the market were excluded. Finally, macro conditioning is partial: while we included FX, inflation, and rates, other structural forces such as capacity constraints or government subsidies were not modelled.

Overall, robustness tests strengthen confidence in our results, but interpretation must remain cautious. These findings should be seen as **credible directional evidence**, not precise forecasts of future returns.

# 8. Recommendations & Next Steps

Our results translate into clear actions for both **airlines** and **investors**.

- **Airlines**:
  - Strengthen **hedging strategies** by tailoring contracts to regional exposures (Brent-linked for Europe/Asia, WTI for the U.S.).
  - Incorporate **sentiment monitoring** into risk management dashboards. A spike in negative sentiment should trigger proactive communication strategies to reassure stakeholders and limit sell-offs.
  - Use macro indicators as an early-warning system. For instance, rising interest rates should automatically raise hedge ratios.
- **Investors**:
  - Factor both **oil price trends and sentiment scores** into allocation models. Simple oil-only screens miss critical behavioural risks.
  - Adopt **regime-based strategies**: overweight airlines in low-oil, stable macro regimes; underweight during inflationary shocks.
  - Employ short-term trading rules around sentiment spikes, particularly for U.S. airlines where the predictive effect is strongest.
- **Next Research Steps**:
  - Apply **machine learning NLP models** (e.g., BERT, FinBERT) to capture richer sentiment signals beyond Bloomberg's proprietary score.
  - Test **options-implied oil volatility** as a leading indicator of airline risk exposure.
  - Expand scope to include **ESG sentiment**, given rising sustainability pressures on airlines.

By integrating these recommendations, both practitioners and researchers can move beyond descriptive correlations and build **actionable playbooks** for navigating energy-linked financial risks.

## 9. Roadmap of the Full Report

The full report expands on these findings with detailed datasets (Section 6.1), comprehensive regression outputs (Section 6.2), volatility and GARCH models (Section 6.3), and event studies of specific airline shocks (Section 6.4). Each team member's contribution provides depth on data sources, methodology, and sectoral interpretation.

## 10. Closing Synthesis

Airline stocks don't just fly on oil — they stall on sentiment.